Ismat Jarin

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Research Interests

Privacy and Machine Learning, Machine Unlearning, Data Privacy and Explainable AI in Virtual Reality, Distributed Machine Learning (FL/Ensemble), Regulation/Policy for AI/ML and LLMs.

Education

Ph.D Candidate, Computer Science, **University of California, Irvine** (2022 – Present)

MS, Computer and Information Science, **University of Michigan, Dearborn** (2019 – 2021)

B.Sc., Electrical Engineering, **Bangladesh University of Engineering and Technology** (2012–2017)

Research Experience

Graduate Research Assistant, ProperData and Networking Group, UCI (2022–present)

Privacy in Virtual Reality:

- Investigating privacy issues in Virtual Reality (VR) associated with Sensor Data Flow, including unique identification and sensitive attribute inference.
- Systematic feature analysis related to privacy exposure in Virtual Reality (VR).

Privacy Preserving Machine Learning:

- Designing audit tool for measuring policy compliance in Machine Unlearning (MU) algorithms, including Deep Learning and Large Language Models (LLMs) services.
- Working on FedMS, a client selection mechanism for Federated Learning (FL) that improves model performance by fairly evaluating rare data clients using Maverick-aware Shapley values.

Visiting Research Assistant, ISI AI Lab, University of Southern California (Summer-2021)

Defense Against Adversarial Patch/Image Manipulation:

- Building a robust deep neural network against adaptive adversarial patch attack. Our goal is to detection, localization and neutralize adversarial patch.

Graduate Research Assistant, University of Michigan-Dearborn (2019–2022)

Privacy Preserving Machine Learning:

- Designed a novel secure Machine Learning pipeline to mitigate Membership Inference Attacks, addressing both probability-dependent and label-dependent vulnerabilities.
- Conducted a comprehensive utility and privacy analysis of Differentially Private algorithms within a machine learning pipeline, evaluating privacy under a membership inference threat model.
- Developed a privacy-preserving machine learning approach using Secure Multiparty Computation and Differential Privacy to safeguard training data for collaborative learning and ensure the privacy of input test data received from external environments or clients.

Research Assistant, Bangladesh University of Engineering and Technology (2017–2018)

- Developed a method for detecting retinal veins from medical images using the ResNet U-net architecture and Fuzzy algorithms.
- Encrypted natural and medical images using self-adaptive permutation and DNA encoding, ensuring secure and efficient transmission of medical images through cryptographic techniques.

Articles in Conference Proceedings

1. **Ismat Jarin**, Tu Le, Yu Duan, Athina Markopoulou "VRProfiLens: Exploration of User Profiling in Real-world Virtual Reality Apps" (ongoing)

2. **Ismat Jarin**, Yu Duan, Rahmadi Trimananda, Hao Cui, Salma Elamiliki, Athina Markopoulou " BehaVR: User Identification Based on VR Sensor Data" Accepted at 25th Privacy Enhancing Technologies Symposium (PETS 2025).
3. Mengwei Yang, **Ismat Jarin**, Baturalp Buyukates, Salman Avestimehr, Athina Markopoulou " Maverick-Aware Shapley Valuation for Client Selection in Federated Learning". Accepted at ISIT 2024 Workshop on Information-Theoretic Methods for Trustworthy Machine Learning.
4. **Ismat Jarin**, Birhanu Eshete "MIAShield: Defending Membership Inference Attacks via Oracle-Guided Exclusion of Members" Accepted at 23rd Privacy Enhancing Technologies Symposium (PETS 2023).
5. **Ismat Jarin**, Birhanu Eshete "DP-UTIL: Comprehensive Utility Analysis of Differential Privacy in Machine Learning" ACM CODASPY 2022.
6. **Ismat Jarin**, Birhanu Eshete "PRICURE: Privacy-Preserving Collaborative Inference in a Multi-Party Setting" *IWSPA '21: Proceedings of the 2021 ACM Workshop on Security and Privacy Analytics*, 2021.
7. **Ismat Jarin**, Tahsin Mostafiz, Celia Shahnaz "Retinal Vein detection using Residual Block Incorporated U-Net Architecture and Fuzzy Inference System" *IEEE International WIE Conference on Electrical and Computer Engineering (WIECON-ECE)*, 2018.
8. **Ismat Jarin**, Celia Shahnaz "Natural and Medical Image Encryption using Self-Adaptive permutation and DNA Encoding."IEEE *IEEE International WIE Conference on Electrical and Computer Engineering (WIECON-ECE)*, 2018.
9. **Ismat Jarin**, Ruhi Sharmin, and Satya Prasad Majumdar , "Performance Evaluation of SISO OFDM System in the Presence of CFO, Timing Jitter and Phase Noise for Rayleigh and Rician Fading Channels." *IEEE Region 10 Humanitarian Technology Conference (R10HTC*, 2017.
10. **Ismat Jarin**, and Satya Prasad Majumdar , "Performance Evaluation of MLD and ZFE in a MIMO OFDM System Impaired by Carrier Frequency Offset, Phase Noise and Timing Jitter" *IEEE International Conference on Telecommunications and Photonics (ICTP)*, 2017.

Teaching and Mentorship Experience

- **Mentor**, UCI Next Generation Pathways (2024 – present).
- **Mentor**, WiCyS Mentorship Cohort (2024 – present).
- **Teaching Assistant**, Computer Science, **University of California, Irvine** (2022 – 2023).
- **Graduate Student Instructor**, Computer Science, **University of Michigan, Dearborn** (2019 – 2022).

Presentations and Invited Talks

- Poster Presentation:*"MLSherlock: An Audit Lens for Policy-Compliance Machine Learning Systems"*, WiML, co-located with NeurIPS'24.
- Poster Presentation:*"BehaVR: User Identification Based on VR Sensor Data"*, UCI AI & Cloud Workshop'24.
- Poster Presentation:*"BehaVR: User Identification Based on VR Sensor Data"*, UCI-UCLA Privacy Workshop'24.
- Invited Penalist: at ProperData IoT Workshop for Undergraduate Students, 2023.
- Paper Presentation: *"MIAShield: Defending Membership Inference Attacks via Preemptive Exclusion of Members"* at PETS'23.
- Paper Presentation: *"DP-UTIL: Comprehensive Utility Analysis of Differential Privacy in Machine Learning"* at ACM CODASPY'22.
- Paper Presentation: *"PRICURE: Privacy-Preserving Collaborative Inference in a Multi-Party Setting"* at IWSPA '21: ACM Workshop on Security and Privacy Analytics.
- Paper Presentation: *"Retinal Vein Detection using Residual Block Incorporated UNet Architecture and Fuzzy Inference System"*, IEEE International WIE Conference on Electrical and Computer Engineering (WIECON-ECE)'18.
- Paper Presentation: *"Natural and Medical Image Encryption using Self-Adaptive Permutation and DNA Encoding"* at IEEE International WIE Conference on Electrical and Computer Engineering (WIECON-ECE)'18.
- Paper Presentation: *"Performance Evaluation of SISO OFDM System in the Presence of CFO, Timing Jitter, and Phase Noise for Rayleigh and Rician Fading Channels"* at IEEE Region 10 Humanitarian Technology Conference (R10HTC)'17.

- Paper Presentation: *"Performance Evaluation of MLD and ZFE in a MIMO OFDM System Impaired by Carrier Frequency Offset, Phase Noise, and Timing Jitter"* at IEEE International Conference on Telecommunications and Photonics (ICTP)'17.

Technical Skills

- **Programming Languages:** Python (proficient), R, C, C++, SQL.
- **Simulation and Design Tools:** Anaconda, Eclipse, MYSQL, MATLAB.
- **Others:** Pytorch, Tensorflow, OpenCV, nltk, Ubuntu/Linuxmint .

Relevant Coursework

Privacy and security, Data Privacy, Network Security, Generative Neural Nets (Audit), Machine Learning, Data Mining,Image Understanding , Algorithm Design, Embedded System, Research in HCI, AI Frontier.

Honors and Awards

- NSF Travel Grant to present ML-Unlearning poster at WiML Workshop, co-located with NeurIPS'24.
- Full funding graduate scholarship from University of California, Irvine.
- Full funding graduate scholarship from University of Michigan.
- WiCyS Security Training Scholarship, 2024.
- Scholarship winner for Grad Cohort Workshop for Women (CRA-WP), 2024.
- Idea contest champion team on ProPerdata Symposium, 2022.
- 4th in Regional round of National Robotics Festival (NRF) in National Robotics Festival, 2014.
- 5th in international IEEE Smarter Planet Challenge Student Competition (December 2013), value $1000.
- ranked 147th to attend Bangladesh University of Engineering and Technology (within top 2% among selected competitors).

Volunteering and Professional Services

- Artifact reviewer committee, USENIX 2025.
- Artifact reviewer committee, PETS 2025, Issue 1, Issue 2.
- Student Volunteer, WiML Workshop, co-located with NeurIPS 2024.
- Student Volunteer, Twenty-third ACM Workshop on Hot Topics in Networks (HotNets), 2024.
- Reviewer for WiML Workshop, co-located with NeurIPS 2024.
- Secretary, WiCyS (Women in CyberSecurity), UCI student chapter, 2023–2024.
- Official Photographer, IEEE ex-com, IEEE BUET student branch, 2015–2017.
- Organizer, IEEE International WIE Conference (2016)
- Organizer, IEEE workshops, seminars, and industrial tours for IEEE BUET Student Branch and WIE BUET Affinity Group (2015–2017).